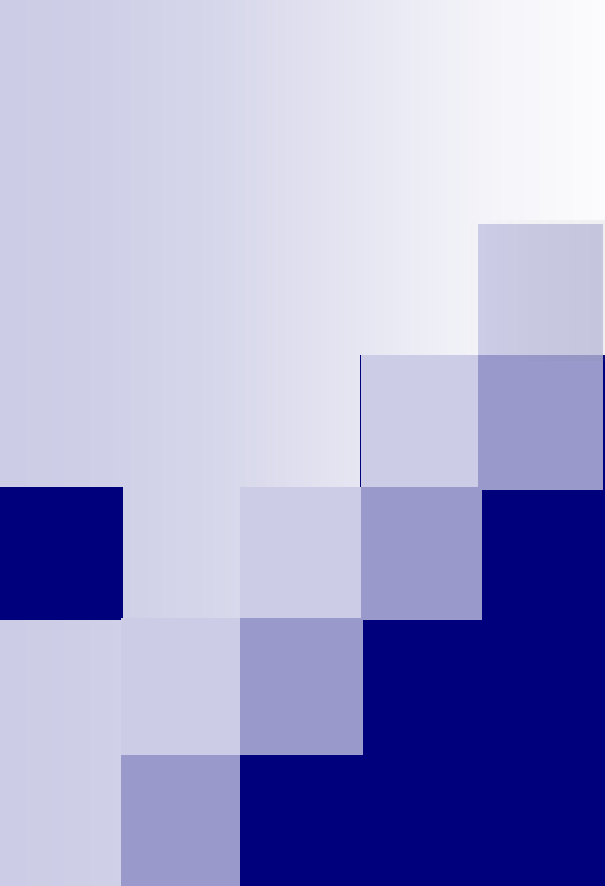




Matteo Palmonari

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5.1

Semantic Data Integration and Information Extraction: Introduction



ITIS Lab – Innovative Technologies for Interaction and Services



*Dipartimento di Informatica, Sistemistica e Comunicazione
Università degli Studi di Milano-Bicocca*

SOME PREMISES...

KBs and Semantic Data Integration

- We can extend the concept of knowledge base
 - Symbolic AI: a set of axioms (in a formal language)
 - facts about individual + rules/constraints that model general knowledge
 - Procedures to manipulate knowledge (edit, query, reasoning, etc.)
 - Key elements from the KG abstraction
 - entities as first-class citizens
 - relations across entities
 - links interconnecting information about entities
 - conceptual modeling of domain entities, e.g., via ontologies
 - The extended interpretation of KB
 - An organized collection of interconnected data and documents
 - KG as abstraction to interconnect data from heterogenous sources (e.g., graphs, tables, texts, images, etc.)
 - Core parts of this collection modeled as full-fledged KGs
- Semantic data integration: ~ the task of building an integrated KB from heterogenous data sources

Semantic Data Integration vs. Matching vs. Information Extraction

■ How to find links across datasets?

□ Matching

- Similarly as in record linkage / entity matching as you have seen in the data management score

□ Information extraction

- Especially when linking unstructured data to structured data

■ Let's look at our example (previous lessons)

Semantic Data Integration

Business

FRANCE - PANAMA

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Par Anne Michel, Jérémie Baruch et Maxime Vaudano

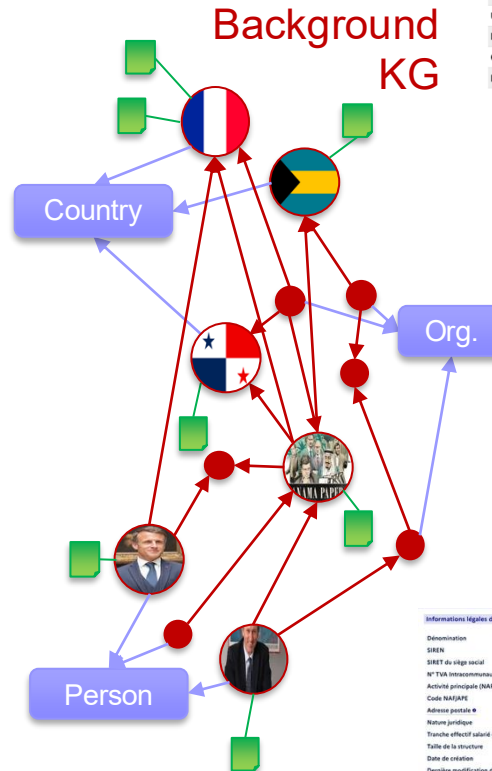
EVASION FISCALE

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En plus d'un redressement fiscal, d'une amende et d'une peine de prison avec sursis, le PDG a dû faire face à des frais non anticipés sur deux sociétés.

Par Jérémie Baruch

Publié le 21 juin 2018 à 10h54, modifié le 22 juin 2018 à 07h02 - Lecture 3 min.



BROWSE BY COUNTRY FRANCE

ENTITY	JURISDICTION	LINKED TO	DATA FROM
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PROCESOS E INGENIERIA S.A.	Panama	France	Panama Papers
WEBCO INTERNATIONAL S.A.	Bahamas	France	Panama Papers
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Entities in OffshoreLeaks linked to France

<https://offshoreleaks.icij.org/search?c=FRA&cat=0>

Foundations firms 'offshore' customers through banks

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Jersey	Société Générale Bank & Trust Luxembourg	465
Luxembourg	Landisbank Luxembourg S.A.	404
Luxembourg	Rothschild Trust Guernsey Limited	378
Guernsey	Banco Santander	119
Spain	BBVA	19

Foundations firms 'offshore' customers through banks in Wikipedia

Informations légales de SIENNA REAL ESTATE HOLDING FRANCE

Dénomination	SIENNA REAL ESTATE HOLDING FRANCE
SIREN	492 220 553
SIRET de siège social	492 220 553 00027
N° TVA Intracommunautaire	FR19 492 220 553
Activité principale (NAJARE)	Fonds de placement et entités financières similaires
Code NAJARE	64.302
Adresse postale	18 RUE DE COUCHELLES, 75008 PARIS 8
Nature juridique	SAS, société par actions simplifiée
Totaux effectif salarié de la structure	Unité non employeuse
Taille de la structure	Petite ou Moyenne Entreprise (PME), en 2020
Date de création	20/09/2006
Dernière modification des données Insee	06/09/2022

Justificatif d'existence

• Télécharger l'extrait RNE (f)

• Avis de situation Insee (f)

Company data from
data.gouv.fr

<https://annuaire-entreprises.data.gouv.fr/entreprise/sienna-real-estate-holding-france-492220553>

Inspiration for this example: [Knoblock&Szekely 2015], [ICIJ](https://www.icij.org/) + Neo4J work for Panama Papers

Semantic Data Integration

Annotations: named entities

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EVASION FISCALE

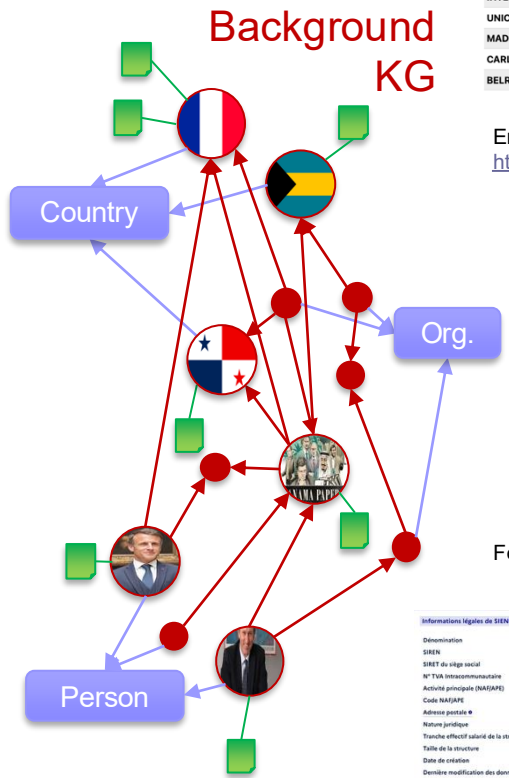
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Named Entity Recognition



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SIENNA REAL ESTATE HOLDING FRANCE

Dénomination : 492 220 553

SIREN : 492 220 553 00027

N° TVA Intracommunautaire : FR19 492 220 553

Activité principale (NAJARE) : 64.30Z

Code NAJARE : 18 RUE DE COURCELLES, 75008 PARIS 8

Nature juridique : SAS, société par actions simplifiée

Toutefois effectif salarié de la structure : Unité non employeuse

Taille de la structure : Petite ou Moyenne Entreprise (PME), en 2020

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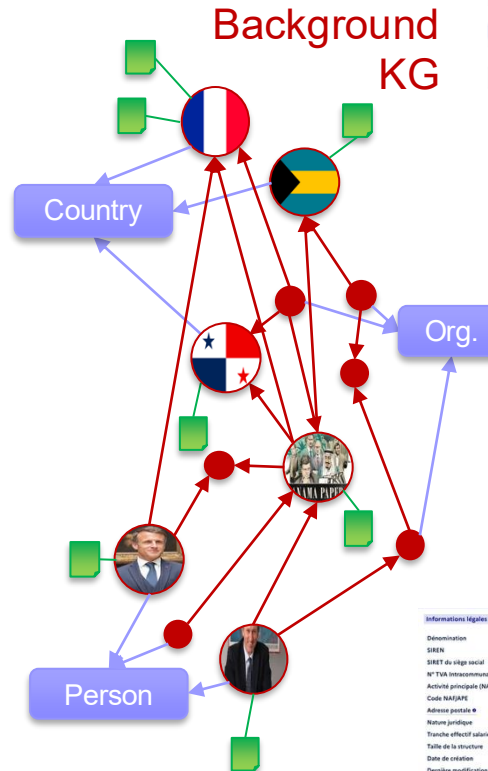
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Named Entity Recognition (NER)



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Semantic Data Integration

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Annotations: data linking

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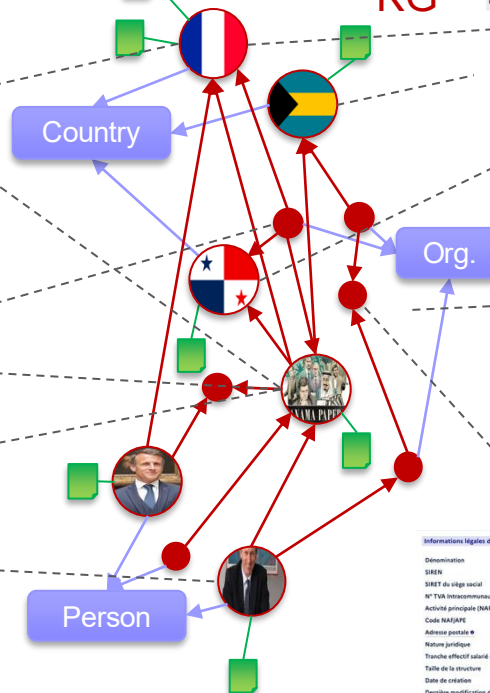
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Background KG



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Semantic Data Integration

[Reminder]

Annotations: data linking

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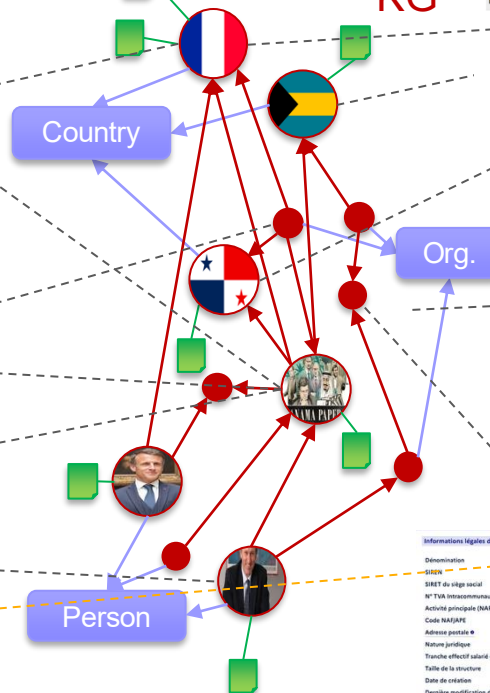
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... Sienna ...

Background KG



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Taux effectif salarié de la structure	Unité sans employeur
Taille de la structure	Personne ou Régime Entreprise (PRE), en 2020
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New entities (not existing in the background KG)

Information Extraction vs. Annotations

■ Information extraction and annotation

□ “extraction”

- Sets of entities, relations, etc.

□ “annotations”

- Links between specific sequences of characters in the text and the extracted content
 - E.g., JSON representing the links
 - E.g., ~ HTML or microformats

Document file

Annotation file

...

Il Generale **Pietro Musumeci** **PERSONA** e il Colonnello **Giuseppe Belmonte** **PERSONA** sono alti ufficiali del **S.I.S.M.I.** **ORGANIZZAZIONE** servizio segreto militare

Nell'estate del 1989 **CATA** l'avvocato di parte civile **Roberto Montorsi** **PERSONA** incontra **Licio Gelli** **PERSONA** e passa dalla parte degli imputati tradendo la fiducia che gli era stata accordata.

Subito si scatena una campagna che cerca di squalificare tutto il lavoro dei magistrati, dell'**Associazione** **ORGANIZZAZIONE** e del **Collegio di Parte Civile** **ORGANIZZAZIONE**.

...

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  "annotation_sets": {
    "sections": {},
    "sentences": {},
    "entities": {
      "annotations": [
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          "type": "PERSON",
          "start": 0,
          "end": 23,
          "id": 0,
          "features": {
            "ner": { "type": "PERSON" },
            "linking": {
              "encoding": "",
              "source": "blink_biencoder",
              "top_candidate": {
                "id": 213062,
                "indexer": 10,
                "wikipedia_id": 534366,
                "title": "Barack Obama",
                "score": 77.57928466796875 },
              "candidates": [],
              "nil_score": 0.43745980781641175,
              "is_nil": true },
            "cluster": 1 }}}],
      "text": "Barack Hussein Obama II (born August 4, 1961) is an ...",
      "features": {}
    }
  }
}
```

Data Integration vs Data Enrichment

■ Data annotation

□ A data enrichment process

- Original data augmented with annotations

□ Support data integration

- Annotations linking to same entity identifiers in different data/documents, support the integration of these data/documents
 - E.g., find all documents that mention “BMP”

THIS INTRODUCTION

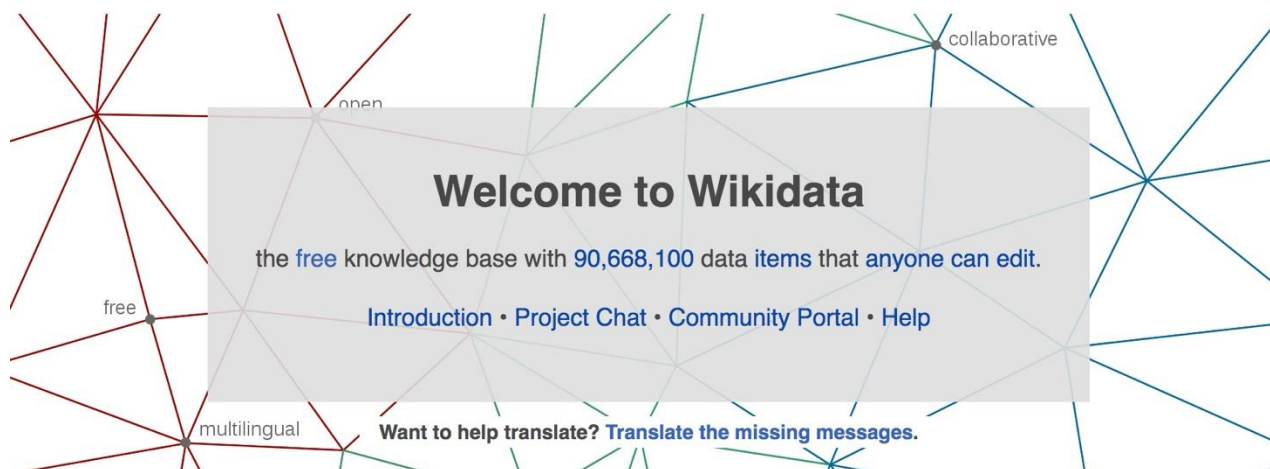
Recap of Previous Premises

- Semantic data integration
 - Can be interpreted as a process where we construct a KB from heterogenous sources
 - Can leverage data enrichment processes
- Three key concepts
 - Matching
 - Information extraction
 - Annotations with links to entity identifiers
- All the above techniques have been revolutionized by LMs

Back to our Question...

■ How to build a Knowledge Base?

- Manual editing does not scale...
- Collaborative efforts:
 - Wikipedia (not as structured as a true KB...)
 - Freebase (acquired by Google and early backbone of Google's KG)
 - Wikidata.org (collaborative but many bots cooperating)



- Most likely, we need to rely on **different kinds of sources** and **transform the information** available therein...

How to Build a Knowledge Base

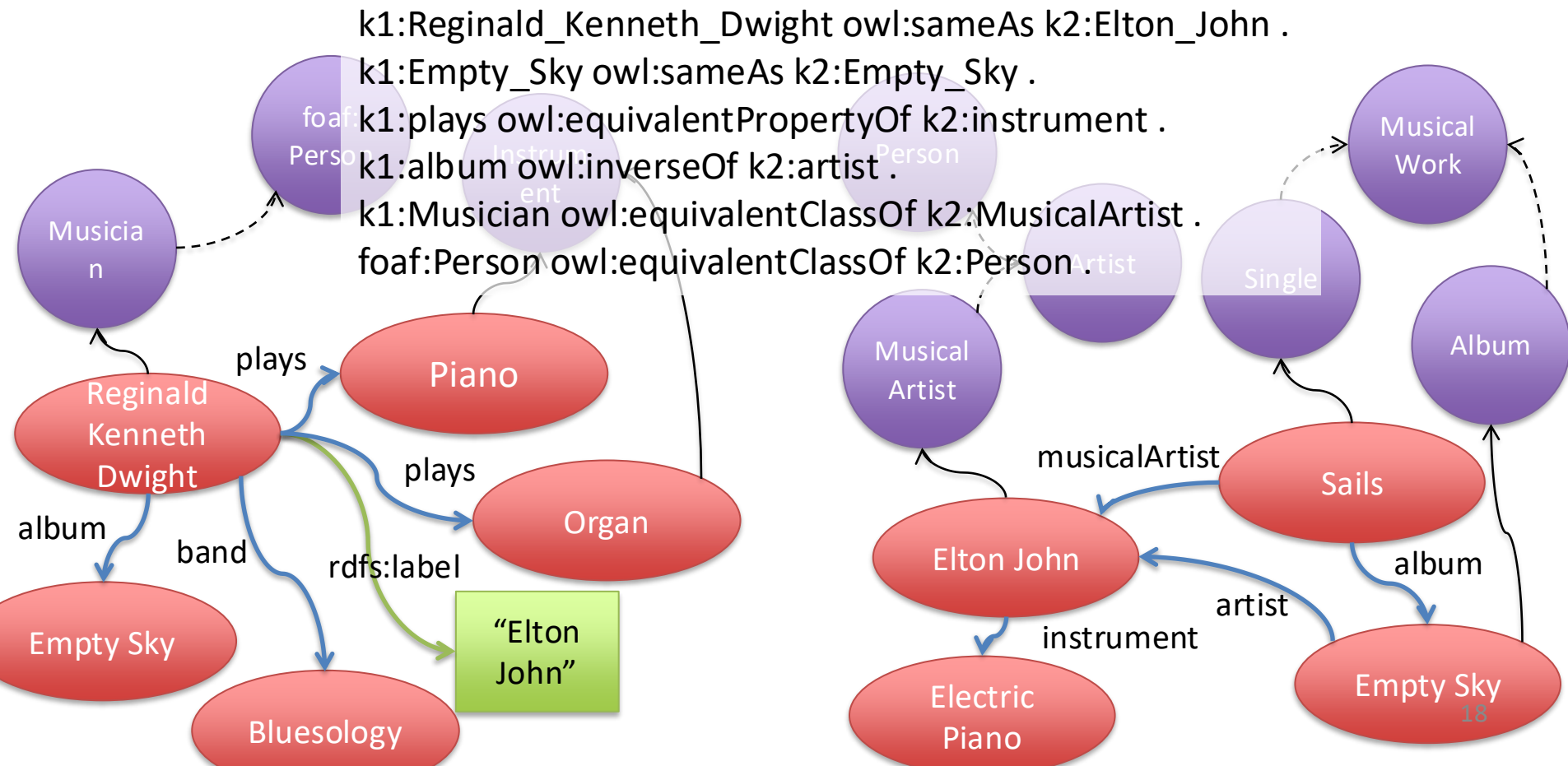
- Often, we need more than one information source
 - which means that data integration (schema and entity matching) is an important component of KB construction
- Sources/techniques we can use:
 - Other KGs / KBs
 - Transformed by mapping & linking
 - Structured sources, e.g., RDB
 - Transformed by mapping, e.g., D2R to map RDBs to RDF, & linking
 - Semi-structured sources, e.g., CSVs, XMLs, etc.
 - Transformed by mapping, e.g., with RML language, & linking
 - Unstructured sources, e.g., texts, images, videos
 - Transformed by information extraction techniques & data integration

How to Build a Knowledge Base

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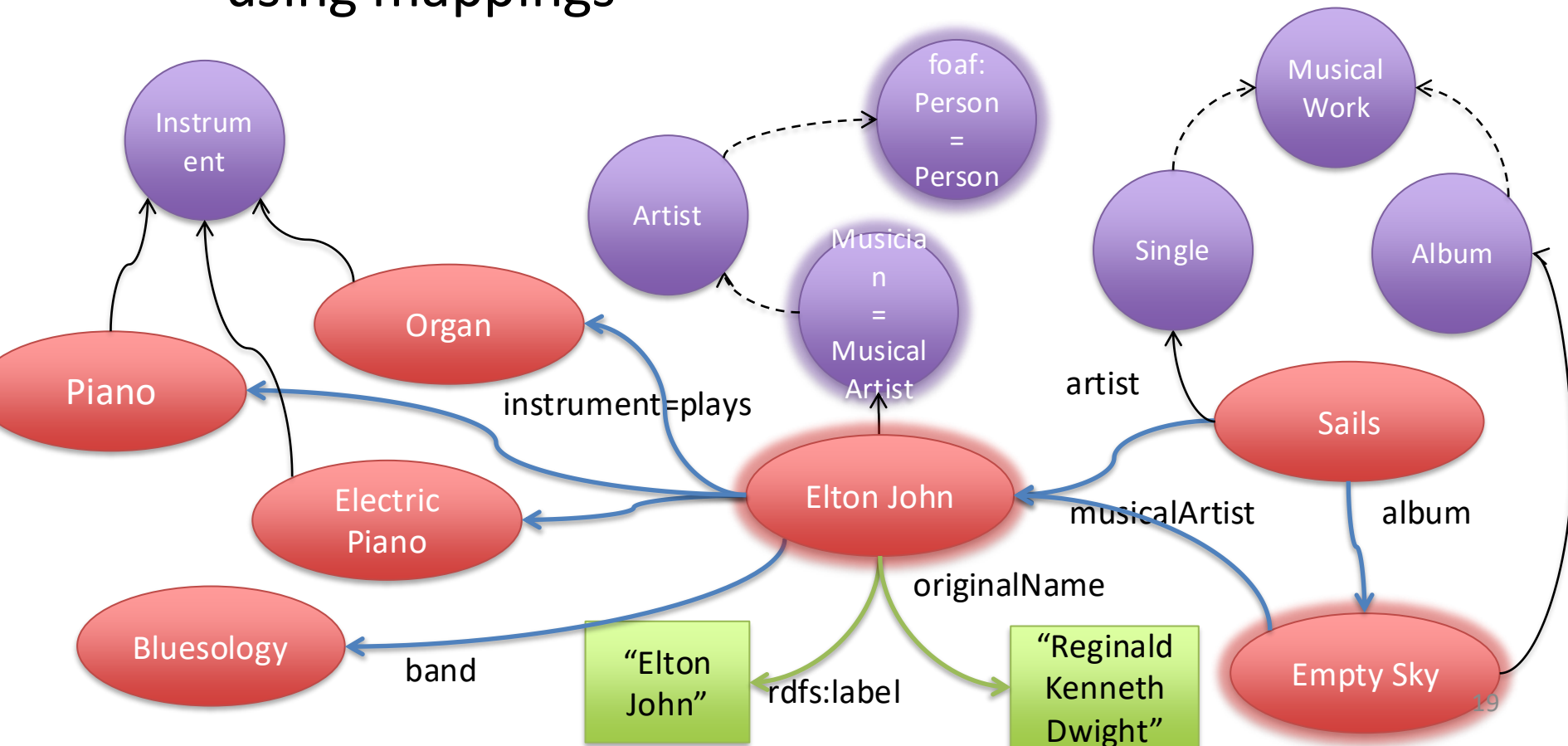
How to build a knowledge base?

- **Align** two KGs so as to make one connected KG
 - Discover and establish mappings, e.g., (simple) links



How to build a knowledge base?

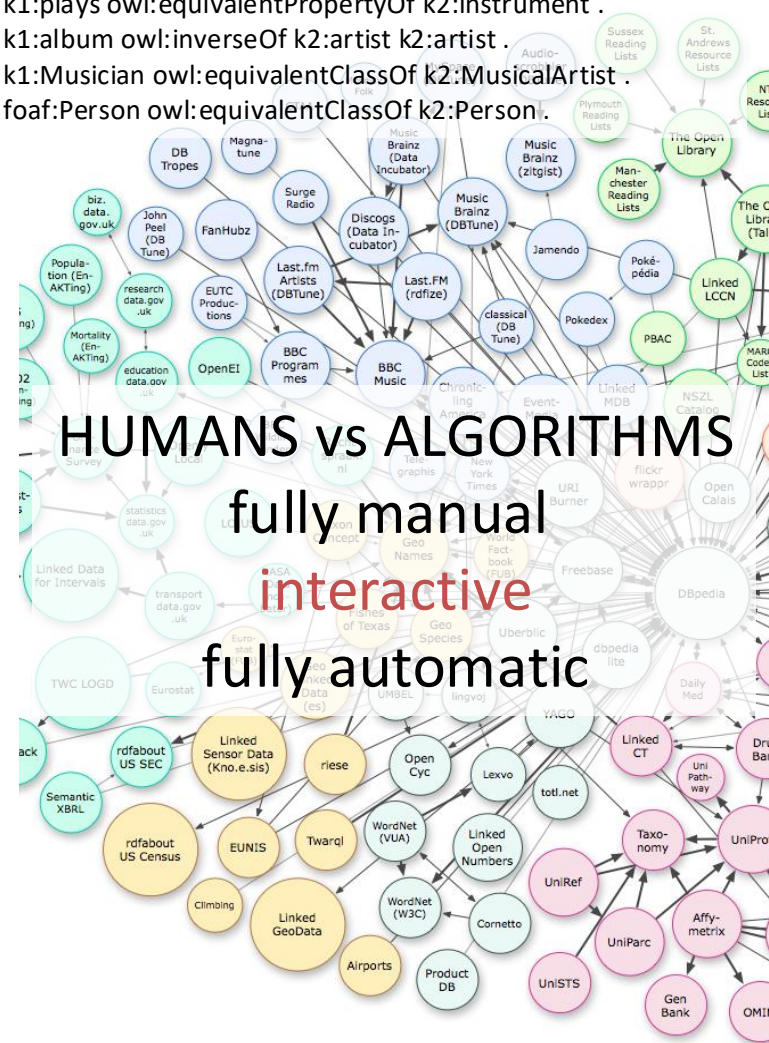
- **Merge** two KGs into one larger KG
 - using mappings



Mapping and Matching

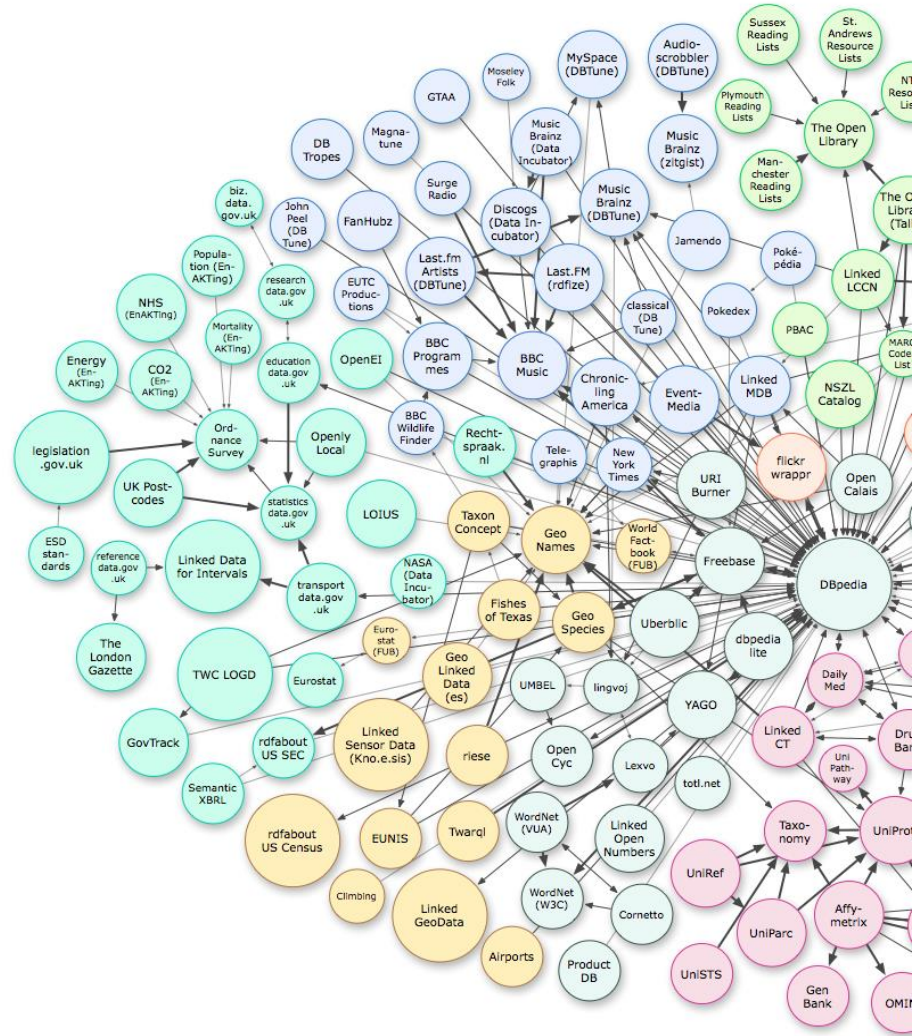
- **Mapping:** the process of specifying
 - which elements of a target dataset correspond to elements of a source dataset
 - (esp. when mapping across heterogeneous models, e.g., table --> KG, see next presentation) how elements from a source dataset are transformed in elements of a target dataset
- **Mappings:** the result of the mapping process
 - the specifications, i.e., the correspondences or the transformations
- **Matching:** the process of finding and deciding about the correspondences
 - focus on the *algorithms* used to automate the mapping process, even when algorithms are mixed with human contributions
- **Matching task:** a specific task of matching two given data sources

k1:Reginald_Kenneth_Dwight owl:sameAs k2:Elton_John .
 k1:Empty_Sky owl:sameAs k2:Empty_Sky .
 k1:plays owl:equivalentPropertyOf k2:instrument .
 k1:album owl:inverseOf k2:artist k2:artist .
 k1:Musician owl:equivalentClassOf k2:MusicalArtist .
 foaf:Person owl:equivalentClassOf k2:Person .



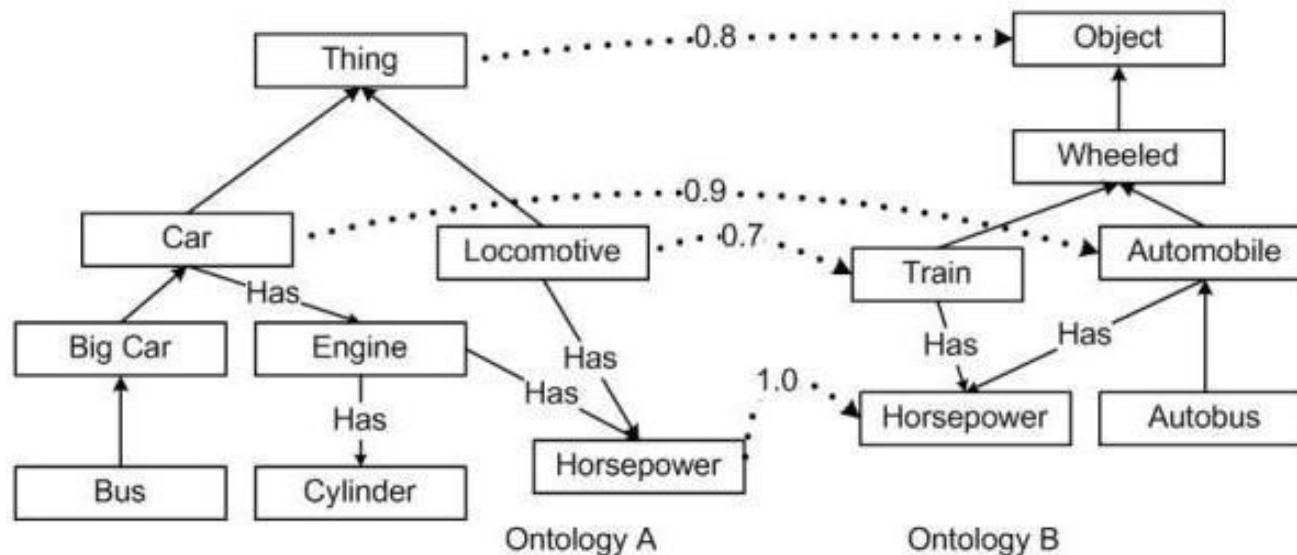
Data Linking across KGs

- Schema-level vs instance-level mappings
 - Schema-level: mappings between concepts and properties
 - **Ontology matching**
 - Instance-level: mappings between entity descriptors in heterogeneous sources
 - **Entity co-resolution** (sameAs)
 - **Instance matching** (sameAs)
 - **Entity matching** (sameAs)
 - **Entity alignment** (sameAs)
 - **Link discovery**
 - $\approx$ Record Linkage (from RDBs)



Ontology Matching

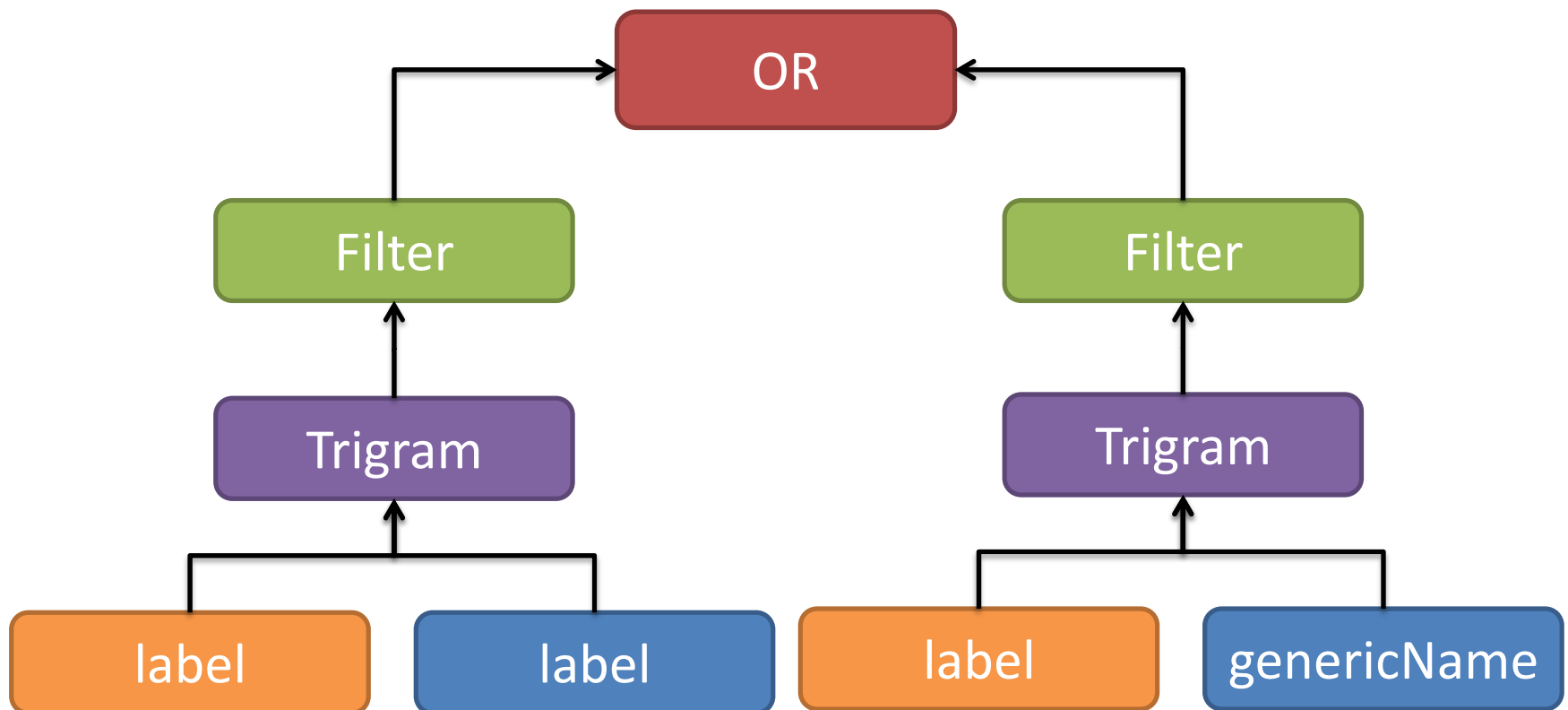
- Finding correspondences (*mappings or agreements*) between entities belonging to different ontologies (*source* and *target* ontologies)
- *mapping* = (*source entity, target entity, relation, similarity*)
- *alignment or matching* = set of *mappings*
- *mapping cardinality*



LIMES Native Interface

Example of instance-level links: “Complex” link specifications learned from a data sample (training data)

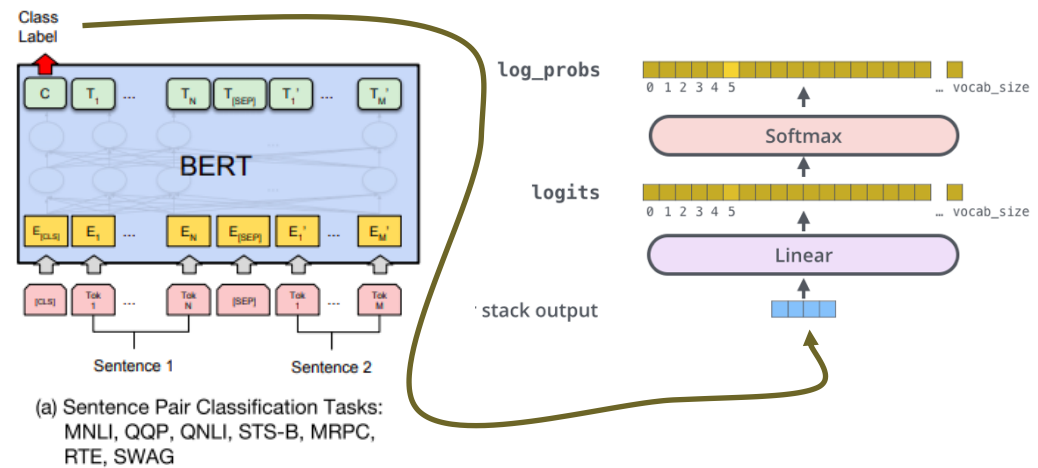
- **OR**(trigram(**x.rdfs:label**, **y.drugbank:genericName**)|**0.8**, trigram(**x.rdfs:label**, **y.rdfs:label**)|**0.8**)



Example from

IMPACT OF LANGUAGE MODELS ON MATCHING TECHNIQUES

DITTO: Deep Entity Matching with Pre-Trained Language Models



- Main Idea
 - Cross-encoder
 - Encode the representations of a pair of candidate entities using the “sentence pair classification” pattern
 - `~~ [CLS] entity1_descr [SEP] entity2_descr`
 - Feed the CLS to a binary classifier (softmax)
 - Return as matches: pairs with TRUE MATCH probability > threshold (typically > 0.5)

Paper (online version: <https://arxiv.org/pdf/2004.00584.pdf>)

DITTO: Deep Entity Matching with Pre-Trained Language Models

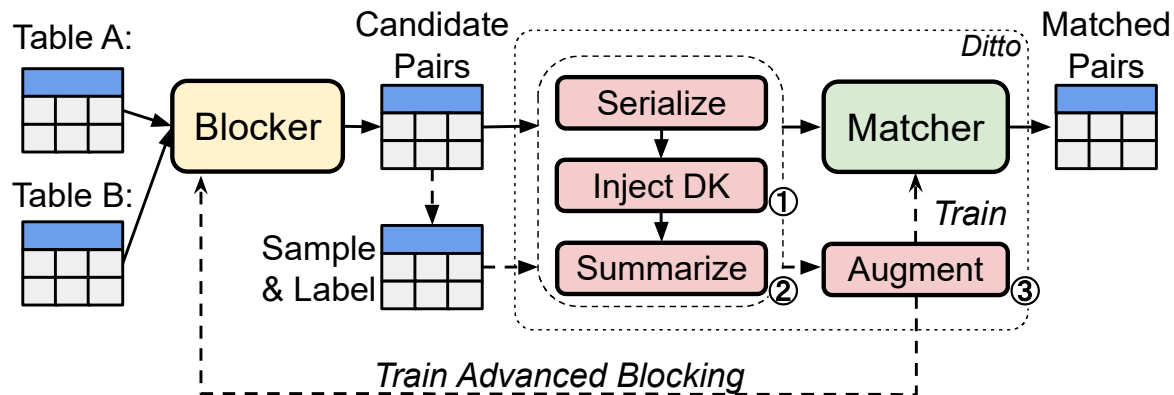


Figure 2: An EM system architecture with DITTO as the matcher. In addition to the training data, the user of DITTO can specify (1) a method for injecting domain knowledge (DK), (2) a summarization module for keeping the essential information, and (3) a data augmentation (DA) operator to strengthen the training set.

- **Serialization**

$\text{serialize}(e) ::= [\text{COL}] \text{attr}_1 [\text{VAL}] \text{val}_1 \dots [\text{COL}] \text{attr}_k [\text{VAL}] \text{val}_k,$

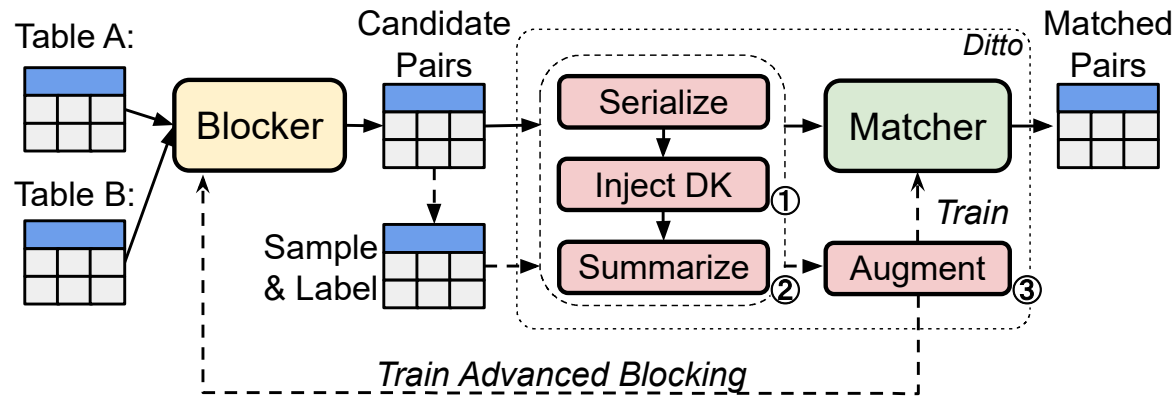
$\text{serialize}(e, e') ::= [\text{CLS}] \text{serialize}(e) [\text{SEP}] \text{serialize}(e')$

- **Other features:**

- Injection of domain knowledge
- Summarization module
- Data augmentation

Paper (online version: <https://arxiv.org/pdf/2004.00584.pdf>)

DITTO: Deep Entity Matching with Pre-Trained Language Models



[COL] title [VAL] instant immers spanish dlux 2 [COL] manf./modelno
[VAL] NULL [COL] price [VAL] 36.11

• Serialization

module for keeping the essential information, and (3) a data augmentation (DA) operator to strengthen the training set.

$\text{serialize}(e) ::= [\text{COL}] \text{attr}_1 [\text{VAL}] \text{val}_1 \dots [\text{COL}] \text{attr}_k [\text{VAL}] \text{val}_k,$

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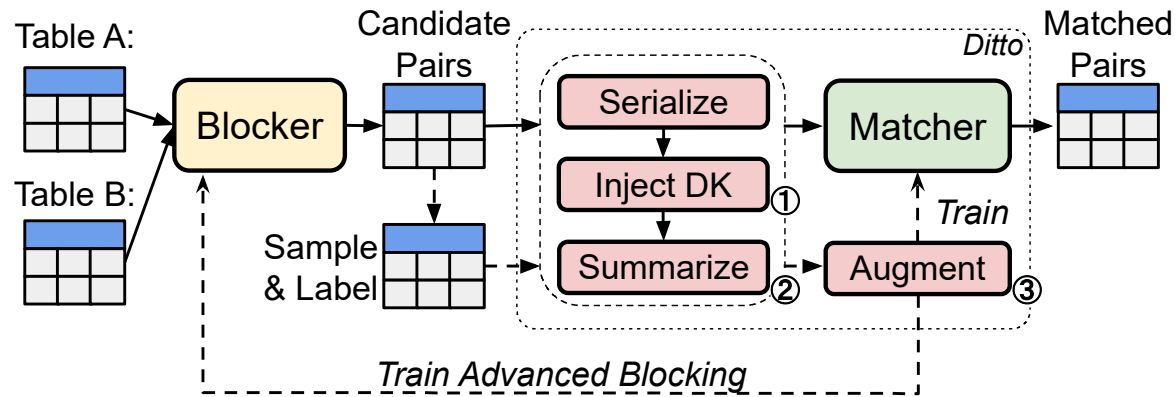


Figure 2: An EM system architecture with DITTO as the matcher. In addition to the training data, the user of DITTO can specify (1) a method for injecting domain knowledge (DK), (2) a summarization module for keeping the essential information, and (3) a data augmentation (DA) operator to strengthen the training set.

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$\text{serialize}(e, e') ::= [\text{CLS}] \text{serialize}(e) [\text{SEP}] \text{serialize}(e')$

Span identification
with NER (e.g., persons) or rules

• Other features:

- Injection of domain knowledge
- Summarization module
- Data augmentation

Entity Type	Types of Important Spans
Publications, Movies, Music	Persons (e.g., Authors), Year, Publisher
Organizations, Employers	Last 4-digit of phone, Street number
Products	Product ID, Brand, Configurations (num.)

E.g., start/end of the last 4 digits for phone numbers
“(866) 246-6453”
→ “(866) 246 - [LAST] 6453 [/LAST]”

Rewriting rules for special spans

E.g., nor commas in integers
e.g., “2,020” → “2020”

DITTO: Deep Entity Matching with Pre-Trained Language Models

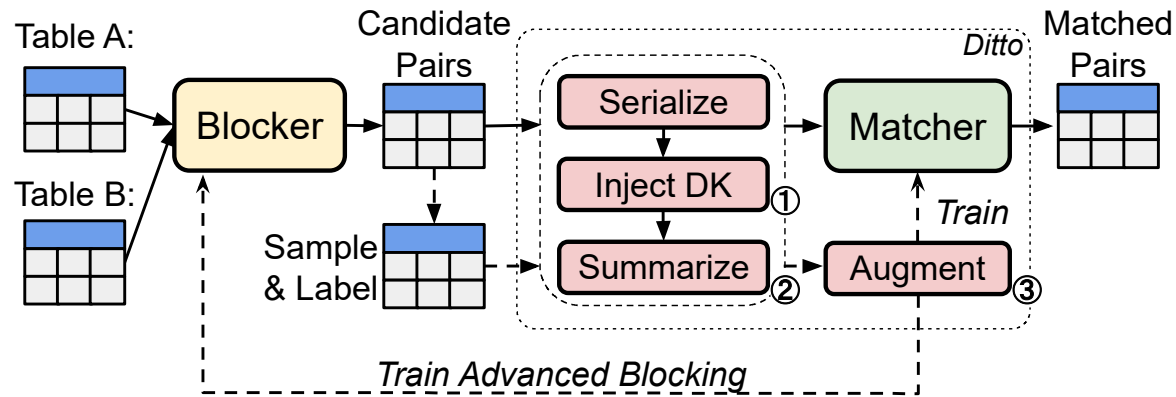


Figure 2: An EM system architecture with D_{ITTO} as the matcher. In addition to the training data, the user of D_{ITTO} can specify (1) a method for injecting domain knowledge (DK), (2) a summarization module for keeping the essential information, and (3) a data augmentation (DA) operator to strengthen the training set.

- **Serialization**

$serialize(e) ::= [COL] attr_1 [VAL] val_1 \dots [COL] attr_k [VAL] val_k,$

$serialize(e, e') ::= [CLS] serialize(e) [SEP] serialize(e')$

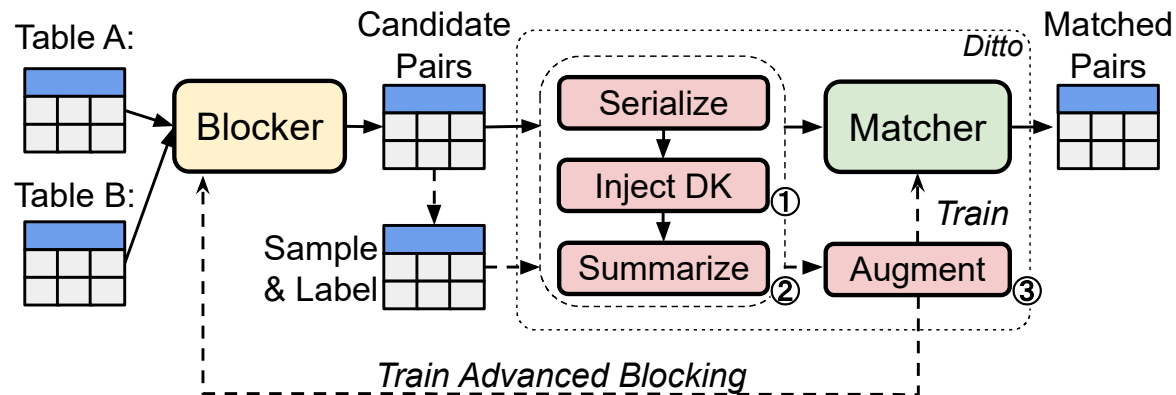
- **Other features:**

- Injection of domain knowledge
- **Summarization module**
- Data augmentation

Retains non-stopword tokens with the high TF-IDF scores for long texts.

...“from 41% F1 to 93% on text-heavy datasets”

DITTO: Deep Entity Matching with Pre-Trained Language Models



[COL] title [VAL] instant immers spanish dlux 2 [COL] manf./modelno
[VAL] NULL [COL] price [VAL] 36.11

- **Serialization**

module for keeping the essential information, and (3) a data augmentation (DA) operator to strengthen the training set.

$\text{serialize}(e) ::= [\text{COL}] \text{attr}_1 [\text{VAL}] \text{val}_1 \dots [\text{COL}] \text{attr}_k [\text{VAL}] \text{val}_k,$

$\text{serialize}(e, e') ::= [\text{CLS}] \text{serialize}(e) [\text{SEP}] \text{serialize}(e')$

- **Other features:**

- Injection of domain knowledge

- Summarization module

- **Data augmentation**

Deletion and reshuffling of some spans / attributes
with interpolation with OG example
Swap entities in entity pairs

Paper (online version: <https://arxiv.org/pdf/2004.00584.pdf>)

A Recent In-depth Study on BERT-based Matching with Useful Insights

PVLDB '22

Analyzing How BERT Performs Entity Matching

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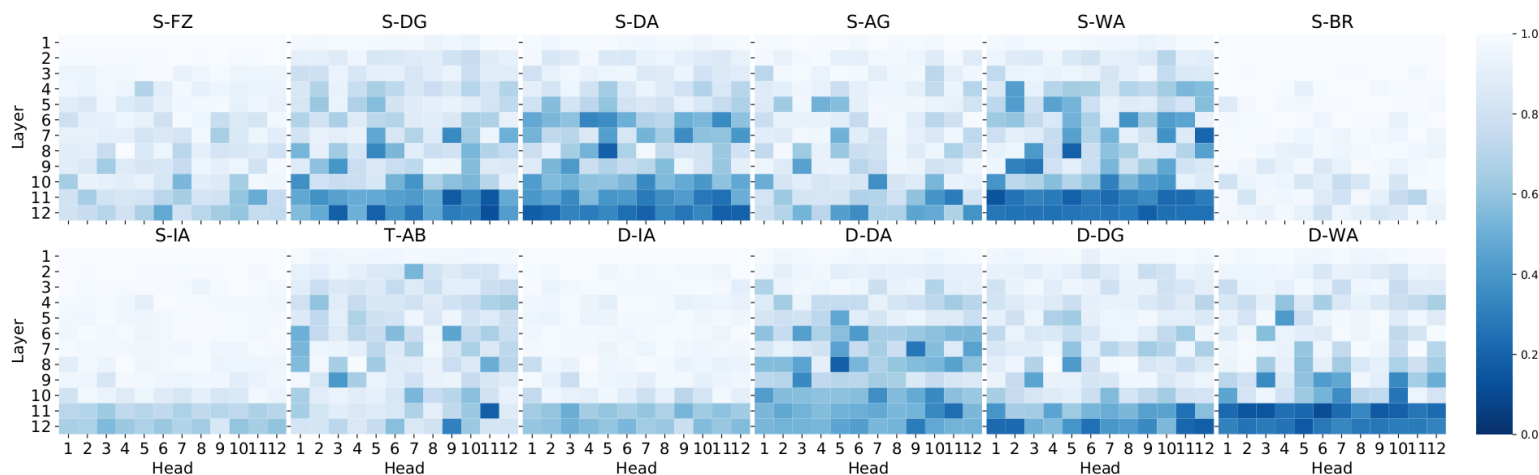
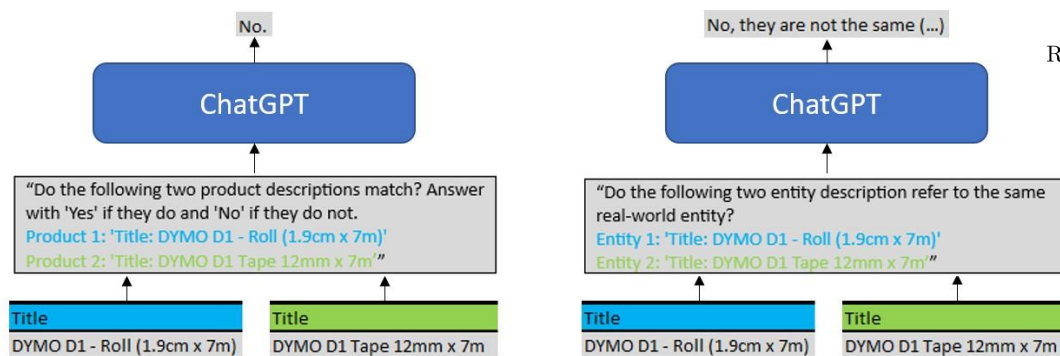


Figure 1: Similarity between pre-trained and fine-tuned attention scores, $settings = (SP, -, PT/FT)$. The darker the cell, the greater the difference between the attention scores of fine-tuned and pre-trained models.

(fine-tuning on binary classification task, not on MLM)

Entity Matching with ChatGPT (Early Results)



Using ChatGPT for Entity Matching

Ralph Peeters^[0000-0003-3174-2616] and Christian Bizer^[0000-0003-2367-0237]

Data and Web Science Group, University of Mannheim, Germany
{ralph.peeters,christian.bizer}@uni-mannheim.de

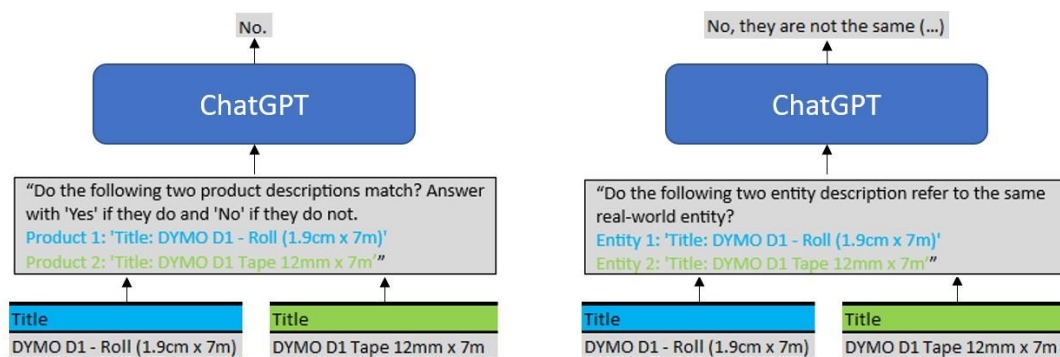
<https://arxiv.org/pdf/2305.03423> (also ADBIS 23)

Published in EDBT25:

<https://openproceedings.org/2025/conf/edbt/paper-81.pdf>

- Pairwise matching = prompt-based entity pair classification (Y | N)
- Domain: products
- Evaluation: performance and costs
- Focus on effects of
 - Different prompts
 - In-context learning (different few-shot strategies)
 - Matching rules in prompt

Entity Matching with ChatGPT (Early Results)



■ Data:

- ❑ Sample of the WDC dataset [\[EDBT 24\]](#)
- ❑ 433 pairs (matches: 11%)
- ❑ Corner cases (difficulty+)
- ❑ SOTA results ~65-89

Table 1. Statistics of the WDC Products benchmark datasets.

Dataset Type	Purpose	# Pairs	# Pos	# Neg
Original Validation	RoBERTa baseline	4,500	500	4,000
Original Training	RoBERTa baseline	19,835	8,741	11,364
Sampled Validation	Evaluation of prompts	433	50	383
Sampled Training	In-context sample selection	2025	898	1,127

Remark: data leakage may require further scrutiny, despite the good will of the authors

Variants

- **General:** These prompts describe the task as the matching of entity descriptions to real-world entities. The product offers are referred to as *entities*. An example of a *general* prompt is the right prompt in Figure 1.
- **Domain:** The domain-specific prompts describe the task as matching of product descriptions and refers to the examples as *product offers*. An example of this type of prompt is the left prompt in Figure 1.
- **Complex:** Prompts in this category use more complex language, specifically they use the formulations "refer to the same real-world product" or "refer to the same real-world entity". An example is the right prompt in Figure 1.
- **Simple:** This type of prompt uses less complex language and replaces the formulations from *Complex* with a simple "match". An example is the left prompt in Figure 1.
- **Free:** This category reflects prompts that do not restrict the models answers in any way. An example is the right prompt in Figure 1.
- **Forced:** In contrast to *Free*, these kinds of prompts explicitly tell the model to answer the stated question with "Yes" and "No". An example is the left prompt in Figure 1.
- **Attributes:** We vary using the three attributes *brand* (B), *title* (T) and *price* (P) in the combinations T, BT and BTP when serializing product offers into single strings.

prompts

few-shot selection

- **Hand-picked:** Hand-picked demonstrations are a set of up to 10 matching and 10 non-matching product offer pairs which were hand-selected by a human domain expert from the pool of the training set.
- **Random:** Demonstrations are drawn randomly from the labeled training set while making sure that they do not contain any of the products that are part of the product offer pair to be matched.
- **Related:** Related demonstrations are selected from the training set by calculating the Jaccard similarity between the pair to be matched and all positive and negative pairs in the training set. The resulting similarity lists are sorted and the most similar examples are selected.

Prompts with rules

Task Desc.	Your task is to decide if two product descriptions match. The following rules regarding product features need to be observed:
Rules	1. The brand of matching products must be the same if available 2. Model names of matching products must be the same if available 3. Model numbers of matching products must be the same if available 4. Additional features of matching products must be the same if available
Task Desc.	Do the following two product descriptions match? Answer with 'Yes' if they do and 'No' if they do not.
Task Input	Product 1: 'Title: DYMO D1 – Roll (1.9cm x 7m)' Product 2: 'Title: DYMO D1 Tape 12mm x 7m'

Results

Table 4. Results of the in-context learning experiments and associated cost.

Selection heuristic	Shots	P	R	F1	Δ F1	Cost (€) per pair	Cost increase	Cost increase per Δ F1
ChatGPT-zeroshot	0	71.01	98.00	82.35	-	0.14	-	-
ChatGPT-random	6	78.33	94.00	85.45	3.10	0.77	450%	145%
	10	79.66	94.00	86.24	3.89	1.13	707%	182%
	20	78.95	90.00	84.11	1.76	2.07	1379%	783%
ChatGPT-handpicked	6	76.19	96.00	84.86	2.51	0.72	414%	165%
	10	80.00	96.00	87.27	4.92	1.00	614%	125%
	20	79.66	94.00	86.24	3.89	2.03	1350%	347%
ChatGPT-related	6	80.36	90.00	84.91	2.56	0.68	386%	151%
	10	89.58	86.00	87.76	5.41	1.05	650%	120%
	20	88.46	92.00	90.20	7.85	1.97	1307%	167%
GPT3.5-handpicked	10	61.97	88.00	72.72	-9.63	10.54	7429%	771%
	20	61.43	86.00	71.67	-10.68	19.71	13979%	1309%
GPT3.5-related	10	67.69	88.00	76.52	-5.83	10.04	7071%	1213%
	20	61.43	86.00	71.67	-10.68	20.34	14429%	1351%

Few-shot selection strategies

Remark: data leakage may require further scrutiny, despite the good will of the authors

+ rules

Prompt	Shots	P	R	F1	Δ F1	Cost (€) per pair	Cost increase	Cost increase per Δ F1
ChatGPT-zeroshot	0	71.01	98.00	82.35	-	0.14	-	-
ChatGPT-zeroshot with rules	0	80.33	98.00	88.29	5.94	0.28	100%	17%
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	10	90.91	80.00	85.11	2.76	1.17	736%	267%
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	20	91.11	82.00	86.32	3.97	2.09	1393%	351%

A follow-up of this paper that performs three main data integration tasks

(<https://arxiv.org/pdf/2603.10547>):

- Schema matching
- Entity matching
- Data fusion

Other Techniques for Entity Matching

■ Graph Neural Networks

- Model relationships in the data
- Especially applied to unsupervised alignment of entities in KGs

■ Multimodal Models

- For multimodal input
 - E.g., product + image

How to Build a Knowledge Base

- Often, we need more than one information source
 - which means that data integration (schema and entity matching) is an important component of KB construction
- Sources we can use:
 - Other KGs / KBs
 - Transformed by mapping & linking
 - **Structured sources, e.g., RDB**
 - Transformed by mapping, e.g., D2R to map RDBs to RDF, & linking
 - Semi-structured sources, e.g., CSVs, XMLs, etc.
 - Transformed by mapping, e.g., with RML language, & linking
 - Unstructured sources, e.g., texts, images, videos
 - Transformed by information extraction techniques & data integration

D2RQ mapping language (D2R)

Check the documentation at <http://d2rq.org/d2rq-language>

```
# CREATE TABLE Conferences (ConfID int, Name text, Location text);
```

```
map:Conference a d2rq:ClassMap;  
  d2rq:dataStorage map:Database1;  
  d2rq:class :Conference;  
  d2rq:uriPattern "http://conferences.org/comp/confno@@Conferences.ConfID@@";  
.
```

```
map:eventTitle a d2rq:PropertyBridge;  
  d2rq:belongsToClassMap map:Conference;  
  d2rq:property :eventTitle;  
  d2rq:column "Conferences.Name";  
  d2rq:datatype xsd:string;  
.
```

```
map:location a d2rq:PropertyBridge;  
  d2rq:belongsToClassMap map:Conference;  
  d2rq:property :location;  
  d2rq:column "Conferences.Location";  
  d2rq:datatype xsd:string;
```

```
# D2RQ Namespace  
@prefix d2rq: <http://www.wiwiss.fu-berlin.de/suhl/bizer/D2RQ/0.1#> .  
# Namespace of the ontology  
@prefix : <http://annotation.semanticweb.org/iswc/iswc.daml#> .  
  
# Namespace of the mapping file; does not appear in mapped data  
@prefix map: <file:///Users/d2r/example.ttl#> .  
  
# Other namespaces  
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .  
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .  
  
map:Database1 a d2rq:Database;  
  d2rq:jdbcDSN "jdbc:mysql://localhost/iswc";  
  d2rq:jdbcDriver "com.mysql.jdbc.Driver";  
  d2rq:username "user";  
  d2rq:password "password";
```

D2RQ mapping language (D2R)

Check the documentation at <http://d2rq.org/d2rq-language>

```
# CREATE TABLE Conferences (ConfID int, Name text, Location text);
```

```
map:Conference a d2rq:ClassMap;  
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  d2rq:class :Conference;  
  d2rq:uriPattern "http://conferences.org/comp/confno@@Conferences.ConfID@@";  
.  
map:eventTitle a d2rq:PropertyBridge;  
  d2rq:belongsToClassMap map:Conference;  
  d2rq:property :eventTitle;  
  d2rq:column "Conferences.Name";  
  d2rq:datatype xsd:string;  
.  
map:location a d2rq:PropertyBridge;  
  d2rq:belongsToClassMap map:Conference;  
  d2rq:property :location;  
  d2rq:column "Conferences.Location";  
  d2rq:datatype xsd:string;
```

See also the RML.io ecosystem for mapping arbitrary semi-structured sources to RDF and generate the data

```
# D2RQ Namespace  
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@prefix map: <file:///Users/d2r/example.ttl#> .  
  
# Other namespaces  
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .  
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .  
  
map:Database1 a d2rq:Database;  
  d2rq:jdbcDSN "jdbc:mysql://localhost/iswc";  
  d2rq:jdbcDriver "com.mysql.jdbc.Driver";  
  d2rq:username "user";  
  d2rq:password "password";
```

LLM x Mapping Construction

KGCW @ ESWC26

<https://openreview.net/pdf?id=qjJF4sIRKj>

Reducing Human Effort in Ontology-Based Data Integration: An LLM-Assisted Mapping Pipeline

Sulthoni Ashiddiqi^{1,*}, Fajar J. Ekaputra^{2,†}, Muhammad Zuhri Catur Candra^{1,†} and Gusti Ayu Putri Saptawati Soekidjo^{1,†}

¹School of Electrical Engineering and Informatics, Institut Teknologi Bandung (SEE-ITB), Bandung, Indonesia

²Institute of Data, Process, and Knowledge Management (DPKM), WU Wien, Vienna, Austria

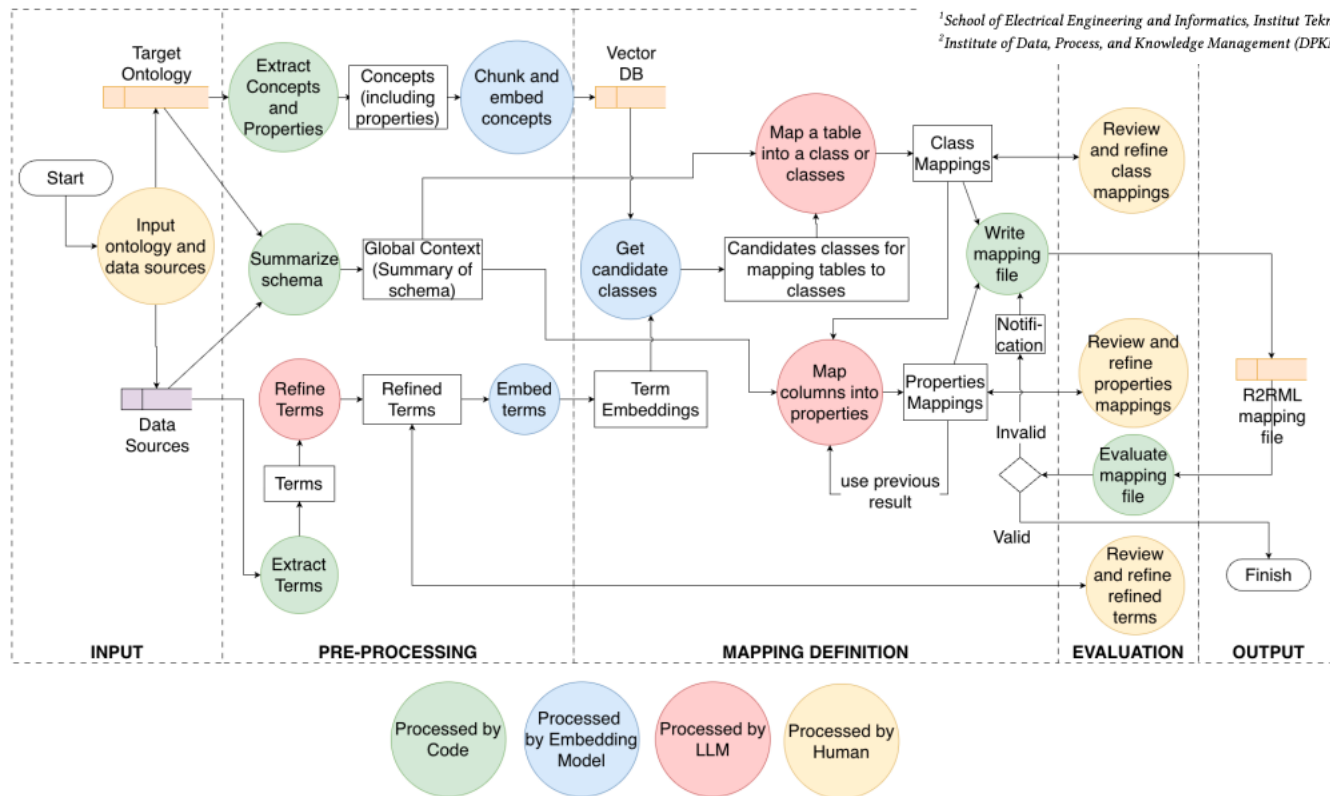


Figure 1: LLM-Assisted Mapping Pipeline Overview. The diagram uses color-coded circles to indicate the category of each process: green circles represent processes executed programmatically; blue circles denote processes supported by an embedding model; red circles indicate processes assisted by an LLM; and yellow circles correspond to processes that involve human input.

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 - **Semi-structured sources**, e.g., CSVs, XMLs, etc.
 - Transformed by mapping, e.g., with RML language, & linking
 - Unstructured sources, e.g., texts, images, videos
 - Transformed by information extraction techniques & data integration

DBpedia – from Infoboxes

- Ontology Design
 - ~730 classes
- Wikipedia infoboxes are mapped to the ontology
 - Triples with “dbo:” properties (~ 1000 OP + 1600 DP) are extracted from these well-curated mappings
- More triples are extracted from the infoboxes, even if properties are not included in the mappings
 - (~ 100k “dbp:” properties!)



<http://mappings.dbpedia.org/>

How to Build a Knowledge Base?

- **Ingest** data from structured or weakly structured sources, e.g., tables

Domain-specific table



Elton John – Empty Sky

Genre: Rock
Style: Pop Rock
Year: 1969

[More Images](#)

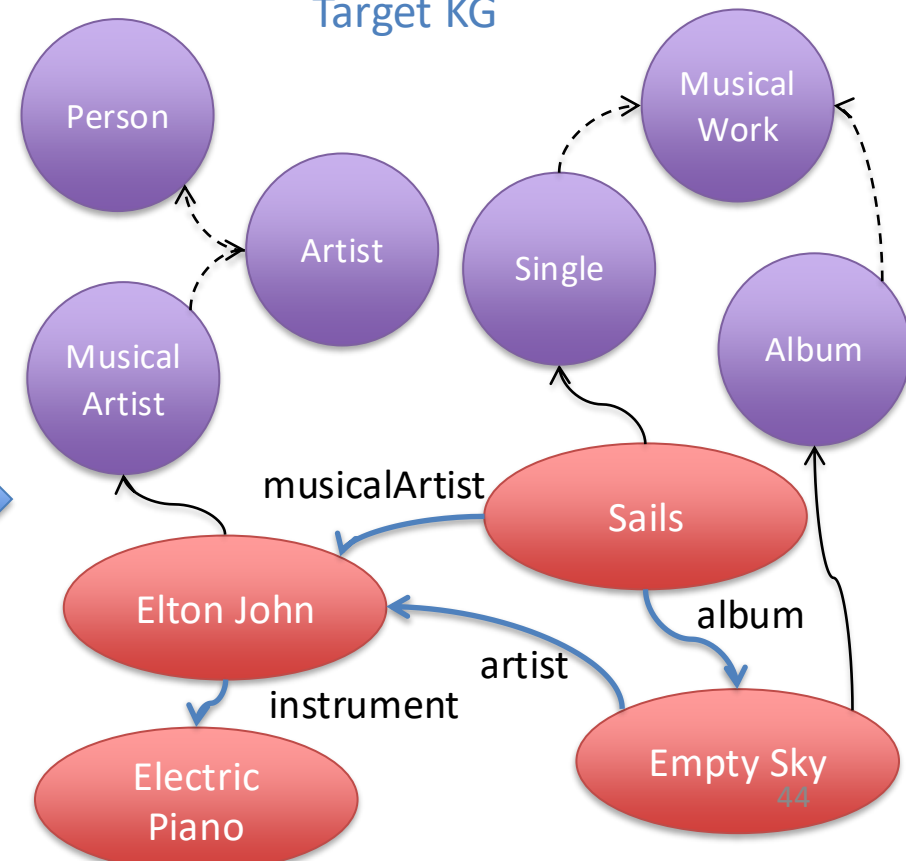
Versions (55)

Q Find Your Version

Title (Format)	Label	Cat#	Country	Year
Empty Sky (LP, Album, Gat)	DJM Records (2), DJM Records (2)	DJLPS 403, DJLPS.403	UK	1969
Empty Sky (LP, Album)	DJM Records (2)	0064 211	Germany	1969
Empty Sky (LP, Album, Mon)	DJM Records (2), DJM Records (2)	DJLPS.403, DJLP 403	UK	1969
Empty Sky (LP, Album, Mono)	DJM Records (2)	DJLP 403	UK & Europe	
Empty Sky (LP, Album, Red)	DJM Records (2), DJM Records (2)	DJLPS 403, DJLPS.403	UK	
Empty Sky (LP, Album)	DJM Records (2), DJM Records (2)	SNIR-DJ 25034, (S/25034/1-/-2)	Italy	
Empty Sky (LP, Album, Gat)	DJM Records (2)	SDJL 934074	New Zealand	

Enrich the target KG with the data from the table

Target KG



How to Build a Knowledge Base

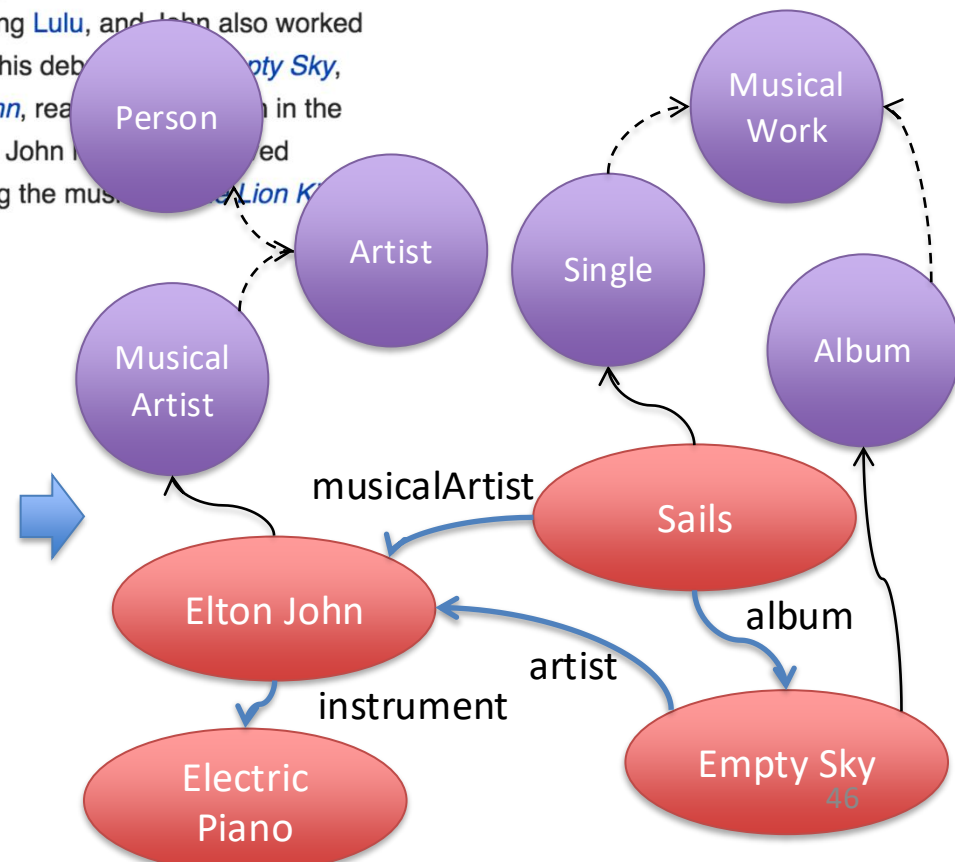
- Often, we need more than one information source
 - which means that data integration (schema mapping or adaptation + entity linking) is an important component of KB construction
- Sources we can use:
 - Other KGs / KBs
 - Transformed by mapping & linking
 - Structured sources, e.g., RDB
 - Transformed by mapping, e.g., D2R to map RDBs to RDF, & linking
 - Semi-structured sources, e.g., CSVs, XMLs, etc.
 - Transformed by mapping, e.g., with RML language, & linking
 - **Unstructured sources**, e.g., texts, images, videos
 - Transformed by information extraction techniques & data integration

How to Build a Knowledge Base?

- **Ingest** data from texts (long or short)

Raised in the [Pinner](#) area of London, John learned to play piano at an early age, and by 1962 had formed [Bluesology](#). John met his songwriting partner, Bernie Taupin, in 1967, after they had both answered an advert for songwriters. For two years they wrote songs for other artists, including [Lulu](#), and John also worked as a session musician for artists such as [the Hollies](#) and [the Scaffold](#). In 1969 his debut album, [Empty Sky](#), was released. In 1970 a single, "[Your Song](#)", from his second album, [Elton John](#), reached number one in the UK and the US, his first hit single. After decades of commercial chart success, John has achieved success in musical theatre, both in the [West End](#) and on [Broadway](#), composing the music for [The Lion King](#) (film and musical), [Aida](#) and [Billy Elliot the Musical](#).

("Elton John") "raised in" "Pinner"
"Pinner" "area of" "London"
("Elton John") "learn to play" "piano"
...



Named Entity Recognition & Linking

WIKIPEDIA
The Free Encyclopedia

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Coldplay

From Wikipedia, the free encyclopedia

Coldplay are an English [alternative rock](#) band formed in 1996 by lead vocalist [Chris Martin](#) and lead guitarist [Jonny Buckland](#) at [University College London](#).^[1] After forming Pectoralz, [Guy Berryman](#) joined the group as a bassist and they changed their name to Starfish.^[2] [Will Champion](#) joined as a drummer, backing vocalist, and multi-instrumentalist, completing the lineup. Manager [Phil Harvey](#) is often considered an unofficial fifth member.^[3] The band renamed themselves "[Coldplay](#)" in 1998,^[4] before recording and releasing three EPs; *Safety* in 1998, *Brothers & Sisters* as a single in 1999 and *The Blue Room* in the same year. The latter was their first release on a major label, after signing to [Parlophone](#).^[5]

The early material of the band was compared to acts such as [Radiohead](#), [U2](#), [A-ha](#), and [Travis](#).^[6] They achieved worldwide fame with the release of the single "Yellow" in 2000, which was nominated for the [Mercury Prize](#). The band's second album, *Parachutes* (2002), was released to favourable reviews and won multiple awards, including *NME*'s Album of the Year. Their next release, *X&Y*, was initially met with mixed reviews. Their fourth album, *Viva la Vida or Death and All His Friends* (2008), was a critical and commercial success, earning several nominations and wins.^[7]

Parlophone and Nettwerk :: Company

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Search in Technorati

Coldplay



Coldplay at the very end of a concert in December 2008. From left to right: [Guy](#)

(Clear Forest Gnosis Mozilla Plugin)

Named Entity Recognition (NER)

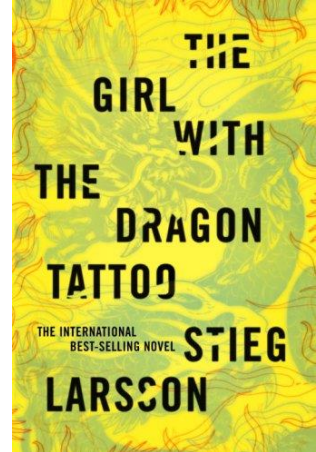
- Extraction of named entities mentioned in a text and classification by type (e.g., person, company, place, etc)

Named Entity Linking (NEL)

- Linking of named entities mentioned in a text to entities described in a KB (e.g., Coldplay is linked to dbr:Coldplay)

Relation Extraction: extraction of facts

(e.g., *machine reading*)



It's about the disappearance forty years ago of Harriet Vanger, a young scion of one of the wealthiest families in Sweden, and about her uncle, determined to know the truth about what he believes was her murder.

Blomkvist visits Henrik Vanger at his home on the island of Hedeby. The old man convinces Blomkvist in by promising solid evidence against Wennerström. Blomkvist agrees to spend a year writing the Vanger family history as a cover for the real assignment: the disappearance of Vanger's niece Harriet 40 years earlier. Hedeby is home to several generations of Vangers, all part owners in Vanger Enterprises. Blomkvist becomes acquainted with members of the extended family, most of whom resent his presence. He does, however, start a short lived affair with Cecilia, the niece of Henrik.

After discovering that Salander has hacked into his computer, Blomkvist hires her to help him with research. They eventually uncover the truth about the disappearance, but Blomkvist has trouble getting close to Lisbeth who treats virtually everyone she meets with hostility. Ultimately the two discover that Harriet's brother Martin, CEO of Vanger Industries, is secretly a serial killer.

A 24-year-old computer hacker sporting an assortment of tattoos and body piercings supports herself by doing deep background investigation. She hires a man named Arman, who, in turn, worries that Lisbeth Salander is "the perfect victim for anyone who wished her ill."

NER, NEL & Relation Extraction

- Different techniques
 - NLP with deep parsing (e.g., syntactic analysis of sentences)
 - Vocabulary-based (e.g., match against a KB)
 - Traditional machine learning (e.g., conditional random fields)
 - LM-based machine learning (e.g., input encoding + classifier or generative)
- Open Information Extraction vs. Closed Information Extraction
 - extraction of any entity or relation (e.g., no dictionary constraints) vs. extraction of entities and relations considering reference lists (e.g., entities of a specific KG or closed set of relation labels)
- Often combined, also depending on the features of the input text
 - Well-formed natural language (e.g., news articles) vs. ill-formed or colloquial natural language (e.g., chats, conversations, etc)

How to Build a Knowledge Base?

□ ~TextBooks and key references

- Gerhard Weikum, Luna Dong, Simon Razniewski, Fabian Suchanek. Machine Knowledge: Creation and Curation of Comprehensive KnowledgeBases. arXiv:2009.11564 [cs.AI] 22 Mar 2021 - <https://arxiv.org/pdf/2009.11564.pdf>
- Kejriwal, Mayank, Craig A. Knoblock, and Pedro Szekely. *Knowledge Graphs: Fundamentals, Techniques, and Applications*. MIT Press, 2021.



■ Tutorials and papers:

- Excellent tutorial from Gabrilovich (Google) and Usunier (Facebook) about web-scale KGs
 - Constructing and Mining Web-scale Knowledge Graphs
 - Slides at <http://www.cs.technion.ac.il/~gabr/publications/papers/SIGIR-2016-KG-tutorial.pdf>
- UCSB group
 - Szekely et al.: Building and Using a Knowledge Graph to Combat Human Trafficking. International Semantic Web Conference (2) 2015: 205-221
 - Craig A. Knoblock, Pedro A. Szekely: Exploiting Semantics for Big Data Integration. AI Magazine 36(1): 25-38 (2015)

We'll Dig Deeper Into

- Information extraction from texts
 - Named Entity Recognition and Linking (details)
 - Relation Extraction (some hints)
- (... if we have time, intro to)
- Information extraction from tables
 - Table interpretation and annotation (web tables, CSVs, etc.)